

OECD NUCLEAR ENERGY AGENCY

**Nuclear Science Committee
Working Party on Scientific Issues of Reactor Systems**

**OECD/NRC Benchmark based on NUPEC PWR
Sub-channel and Bundle Tests (PSBT)
Second Workshop (PSBT-2)**

SUMMARY RECORD

**11-12 April 2011
Royal Institute of Technology (Kungliga Tekniska Högskolan, KTH)
Stockholm, Sweden**

Jim GULLIFORD Jim.GULLIFORD@oecd.org +33 1 45 24 10 72

JT03307416

Document complet disponible sur OLIS dans son format d'origine
Complete document available on OLIS in its original format



OECD/NEA Nuclear Science Committee
and
Committee on The Safety of Nuclear Installations

**OECD/NRC Benchmark based on NUPEC PWR
Sub-channel and Bundle Tests (PSBT)
Second Workshop (PSBT-2)**

11-12 April 2011
Royal Institute of Technology (Kungliga Tekniska Högskolan, KTH)
Stockholm, Sweden

Summary Record

Sponsorship

1. At the Sixth and last workshop for the OECD/NRC Benchmark based on Nuclear Power Engineering Corporation (NUPEC) Boiling Water Reactor Full-size Fine-mesh Bundle Tests (BFBT-6), which was held on April 27-28, 2009 in University Park / State College, PA, USA, a new OECD benchmark based on NUPEC Pressurized Water Reactor Sub-channel and Bundle Tests (PSBT) was introduced.

2. The OECD/NRC PSBT benchmark activities are being performed as an international project supported by US Nuclear Regulatory Commission (NRC) and Japanese Ministry of Economy, Trade and Industry (METI), and endorsed by the Nuclear Energy Agency of the Organization for Economic Co-operation and Development (OECD/NEA). The benchmark team are organized based on the collaboration between USA and Japan as shown in Figure 1. This approach has been very successful in the OECD/NRC BFBT benchmark, which has been completed. It shall be recognized that METI had sponsored the NUPEC PWR subchannel and bundle test project. Japan Nuclear Energy Safety Organization (JNES) on behalf of METI provides the test information and measured data. OECD/NEA provides supporting activities for the international PSBT benchmark project. The US team, headed by the Pennsylvania State University (PSU), co-ordinates the benchmark project in cooperation with US NRC. The activity of the US benchmark team (PSU) is sponsored by the US NRC.

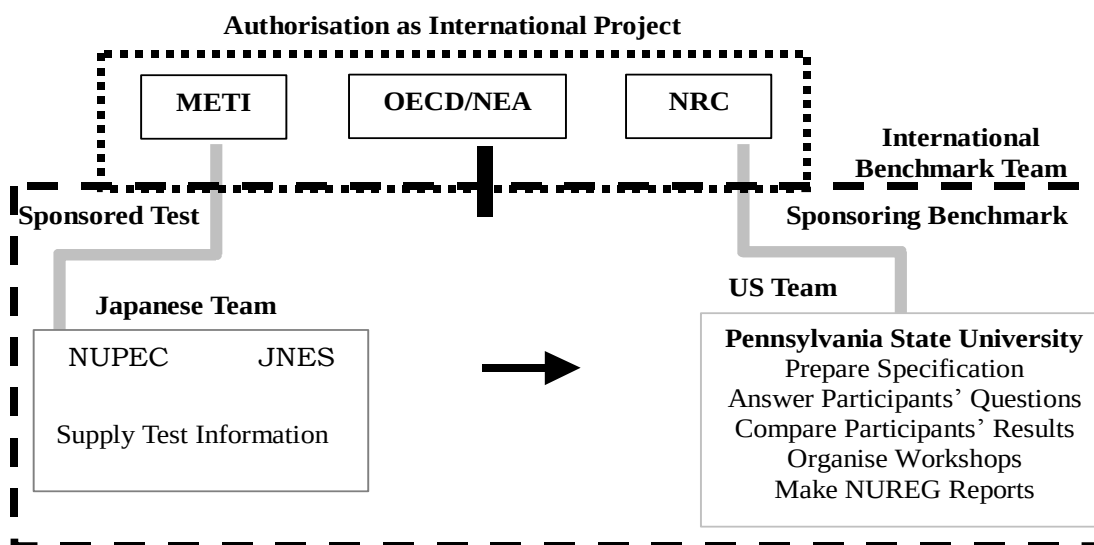


Figure 1. PSBT Benchmark Team

3. This workshop (PSBT-2) was held in conjunction with other meetings, in order to facilitate co-ordination and sharing of work. The four other meetings were held at the same place and during the same week in order to combine efforts in common areas such as Computational Fluid Dynamics (CFD) modelling and uncertainty analysis and to make the participation more efficient. The meetings concerned were the Fifth Workshop for the OECD Uncertainty Analysis in Modelling (UAM) Light Water Reactor (LWR) benchmark (UAM-5), which took place on 13-15 April, and the AER Working Group D workshop (VVER dynamics and safety), which took place on 12-13 April 2011. In parallel with the PSBT-2 meeting, also the Third OECD Kalinin-3 VVER-1000 Benchmark Workshop (K3-3) was held at the same premises. The First workshop of the OECD/NRC Oskarshamn-2 benchmark took place on 13 April 2011.

Background and Purpose of the Benchmark Workshop

4. Over the last few years the Pennsylvania State University under the sponsorship of the US NRC has prepared, organized, conducted and summarized the OECD/NRC Benchmark based on NUPEC BWR Full-size Fine-mesh Bundle Tests (BFBT). The international benchmark activities have been conducted in cooperation with the Nuclear Energy Agency, OECD and Japan Nuclear Energy Safety Organization, Japan. From 1987 to 1995, Nuclear Power Engineering Corporation in Japan performed a series of void measurement tests using full-size mock-up tests for both BWRs and PWRs. For BWRs, based on the state-of-the-art computer tomography (CT) technology, the void distribution was visualized at a mesh size smaller than the subchannel under actual plant conditions. NUPEC also performed steady-state and transient critical power test series based on equivalent full-size mock-ups. Considering the reliability not only of the measured data, but also other relevant parameters such as the system pressure, inlet sub-cooling and rod surface temperature, these test series supplied the first substantial database for the development of truly mechanistic and consistent models for void distribution and boiling transition.

5. Consequently, the JNES has made available the BWR NUPEC database for the purposes of the OECD/NRC BFBT international benchmark. This international benchmark has encouraged advancement in the uninvestigated fields of two-phase flow theory with very important relevance to the nuclear reactors safety margins evaluation. The BFBT benchmark was made up of two parts (phases), each part consisting of four different exercises and was used for validation of CFD, subchannel, and system codes. The BFBT benchmark activity has been very successful with about thirty (30) organizations from fifteen (15) countries participating in different benchmark exercises. The BFBT activity has been completed with final comparative reports to be published as US NRC NUREG and NEA/OECD reports. A special Nuclear Engineering and Design journal issue devoted to the BFBT benchmark, documenting the models and results of the participants, is being finalized.

6. Based on the success of the OECD/NRC BFBT benchmark the JNES, Japan has decided to release also the data based on the NUPEC PWR Subchannel and Bundle Tests (PSBT) for an international benchmark and has asked PSU to organize and conduct this benchmark activity. Void fraction measurements and departure from nucleate boiling (DNB) tests were performed at NUPEC under the conditions simulating PWR thermal-hydraulic conditions, including the steady states and the transients such as the power increase, the flow reduction, the depressurization and the temperature increase.

7. The established international OECD/NRC PWR Subchannel and Bundle Tests (PSBT) benchmark, based on the NUPEC database, encourages advancement in subchannel analysis of fluid flow in rod bundles, which has very important relevance to the nuclear reactor safety margin evaluation. The benchmark specification is designed to systematically assess and compare the participants' numerical models for the prediction of detailed subchannel void distributions and DNB to full scale experimental data on a prototypical PWR rod bundle. Currently the numerical modeling of subchannel void distribution has limited theoretical approach that can be applied to a wide range of geometrical and operating conditions. In the past decade, experimental and computational technologies have tremendously improved the study of the flow structure. Over the next decade, it can be expected that mechanistic approaches will be more

widely applied to the complicated fluid phenomena inside fuel bundles. The development of truly mechanistic models for DNB prediction is currently underway. These models must include processes such as void distribution, critical heat flux, and spacer grid behavior. The benchmark specification requires participants to explain their modeling correlations between the measured DNB and the flow dominant processes.

8. It should be recognized that the purpose of this benchmark is not only the comparison of currently available computational approaches but above-all the encouragement to develop novel next-generation approaches that focus on more microscopic processes. In this context, the subchannel grade void fraction data are regarded as the macroscopic data and the computer graphic images are the microscopic data, which provides the detailed void distribution within the subchannel.

9. There are two separate phases, each consisting of different exercises. These phases and exercises are discussed below.

Phase I - Void Distribution Benchmark

10. The purpose of this benchmark phase is three-fold:

1. to provide data for validation of numerical models of void distribution;
2. to provide data over a wide range of geometrical and operating conditions for validation of void distribution models;
3. to provide data for development of mechanistic approaches widely applicable to the fluid phenomena inside fuel bundles.

11. Phase I includes four exercises:

Exercise 1 – Steady-state single subchannel benchmark, which can be used to improve the current models of void generation and void distribution within a subchannel.

Exercise 2 – Steady-state bundle benchmark, which can be applied to subchannel and system thermal-hydraulic codes as well as CFD codes.

Exercise 3 – Transient bundle benchmark, which can be used to determine the capabilities of subchannel and system thermal-hydraulic codes to predict void fraction during transients.

Exercise 4 – Pressure drop benchmark. Code-to-code comparisons will also be performed.

Phase II - Departure from Nucleate Boiling (DNB) Benchmark

12. The purpose of this benchmark is to develop truly mechanistic models for DNB prediction.

13. Phase II includes three exercises:

Exercise 1 – Steady-state fluid temperature benchmark, which can be applied to thermal-hydraulic subchannel and system codes and CFD codes.

Exercise 2 – Steady-state DNB benchmark, which can be used to improve the current models of DNB prediction in PWR bundles at steady state conditions.

Exercise 3 – Transient DNB benchmark, which can be used to improve the current models of DNB prediction in PWR bundles during transients.

Scope and Technical Content of the PSBT-2 Workshop

14. The technical topics addressed at the workshop included:
- discussion of the modifications and additional data included in the final benchmark specification;
 - discussion of the final results for Exercises 1, 2, 3, and 4 of Phase I;
 - discussion of the input data, modelling issues, and preliminary results for the Exercises 1, 2, and 3 of Phase II;
 - preparing the revised schedule of the benchmark activities.

Organization and Programme Committee of the PSBT-2 Workshop

15. An Organization and Programme Committee has made the necessary arrangements for the Second Benchmark Workshop, organized the Sessions, and prepared the final program. The general chair was Maria Avramova (Coordinator of OECD/NRC PSBT benchmark). The other members were Hideaki Utsuno (JNES), Alexander Velazquez-Lozada (US NRC), Tomasz Kozlowski (KTH, Chair of EGUAM at NSC), representing the host organization, Francesco D'Auria, representing CSNI, Kostadin Ivanov (Coordinator of the OECD LWR UAM benchmark), and Jim Gulliford, representing the OECD/NEA Secretariat.

Opening Session – Introduction and opening remarks

16. The meeting was opened by Prof. Tomas Lefvert, who welcomed the participants on behalf of the host organization and wished them a successful work. Dr. Jim Gulliford presented the structure of working party on scientific issues in reactor systems at the Nuclear Science Committee (NSC) of NEA/OECD, and welcomed the participants on behalf of NEA Secretariat.

17. The agenda was approved with minor adjustments (see *Appendix 1*).

18. The workshop was attended by 41 participants from 23 organizations in 13 countries (see *Appendix 2*). The interest in this benchmark was very large as shown by the participation from research institutions, universities and industry with sub-channel, CFD, porous media and system thermal-hydraulic codes.

19. Dr. Maria Avramova (coordinator of the OECD/NRC PSBT benchmark) discussed the status of PSBT benchmark activities.

Technical Sessions on Phase I – Void Distribution Benchmark

20. Sessions I to V were devoted to the four exercises of Phase I of the PSBT benchmark.

21. Adam Rubin (PSU) made two (2) presentations on “Discussion on modifications and additional data included in the final benchmark specification” and “Presentation of support studies performed by the benchmark team”.

22. The benchmark team summarized, in four (4) presentations, submitted preliminary results for Exercise I-1(single subchannel steady state void distribution), Exercise I-2 (bundle steady state void distribution), Exercise I-3 (bundle transient void distribution), and Exercise I-4 (pressure drop).

23. Participants made eighteen (18) presentations on the physical models of the computer codes used and the final results obtained for the four exercises of Phase I.

24. Based on these presentations and discussed issues, the following decisions were made:

- Void fraction measurements and associated uncertainties:
 - Higher measurement uncertainties to be used for the low void test cases (0 – 20%). This is to account for the fact that the mask applied to the CT data covers some of the near wall flow as well as other “near wall effects” due to the very high void fraction near the wall. Dr. David Novog (McMaster University) proposed the following methodology: First, a raw image for a representative test will be examined and compared to a CAD (or CFD model) to determine what percentage of the cross-section area is missing due to the mask. If we assume 100% vapor in the masked area and 100% liquid for the rest of the flow, the “maximum” void fraction error can be found. This is applicable at the low void fraction levels and should reduce as the flow regime transitions to cases where the void is dominated in the gap areas. It was proposed that this additional void fraction uncertainty is linearly varied from 0% to 15% measured void. An alternative is to “step” change the void uncertainty and have it be 8% void in the region 0 - 20% measured void. The benchmark team will perform more detailed calculations.
 - Higher measurement uncertainties to be used for the high void fraction region. This is to account for the fact that the reported void fraction from the CT measurements is non-symmetric, resulting from small asymmetry in the experiment which cannot be totally eliminated. Dr. David Novog (McMaster University) proposed the following methodology:
 - “Pixel” void fraction (CFD): Determine the void fraction per pixel for the entire subchannel flow area. Then compare the experimental void fraction at each symmetrical pixel and establish a minimum and maximum. Look at the probability distribution function of the minimum and maximum and determine the appropriate pixel by pixel variance due to asymmetry effect. Alternatively, compute the mean and standard deviation and use the deviation as an additional experimental component.
 - Subchannel void fraction: Divide the channel geometry into 4 symmetrical parts. Compare the computed void fraction across the 4 regions to the average void fraction for the entire subchannel. Based on the standard deviation determine an asymmetrical void fraction uncertainty and add it to the CT data.
- Boundary Conditions:
 - The benchmark team will perform a parametric study on the location of the fluid temperature measurements. The fluid temperature measurements were taken in the pipe between the test vessel inlet nozzle and the pre-heater (*Figure 2.2.1, A. Rubin, et al., “OECD/NRC PSBT Benchmark, Volume I: Experimental Database and Final Problem Specifications”, [NEA/NSC/DOC\(2010\)1](#), November 2010*). The pipe size is 100AS160. The pipe length from the measurement location to the vessel coolant inlet is unknown. As a result, a time shift between the experimental data and the codes’ predictions is observed; most significantly in the simulations of the temperature increase transient.
- Spacer Grid Data:
 - CFD participants are encouraged to investigate the spacers’ orientation along the bundle length.
 - The heater rod outer diameter in the provided spacers’ schematics to be confirmed.
- Additional requested output:

- Exercise I-1: All participants to submit calculated void axial profile for the following test cases: 1.2211 (reference case), 1.2223, 1.2237, 1.4326, and 1.4325; and for the bundle configuration B5 of Exercise I-2: 5.2332.

Technical Sessions on Phase II – Departure from Nucleate Boiling Benchmark

25. Sessions VI to IX were devoted to the four exercises of Phase II of the PSBT benchmark.
26. Adam Rubin (PSU) summarized, in three (3) presentations, submitted preliminary results for Exercise II-1 (steady state fluid temperature benchmark), Exercise II-2 (steady state DNB benchmark), and Exercise II-3 (transient DNB benchmark).
27. Participants made nine (9) presentations on the physical models of the computer codes used and the preliminary results obtained for the three exercises of Phase II.
28. Dr. Jean-Marie Le Corre (Westinghouse, Sweden) made two presentations on the applications of the in-house subchannel code VIPRE-W/MEFISTO-T to the critical power and pressure drop exercises of the OECD/NRC BFBT benchmark.

Conclusions, Actions, and Schedule for conducting the PSBT benchmark

29. The action items and schedule for the future PSBT activities were discussed. They are provided in the following list.

List of Agreed Actions on PSBT

1. End of June 2011 - Final results for Phase I.
 2. End of August 2011 - Final results for Phase II.
 3. September 25, 2011 - The PSBT-3 workshop will be held in conjunction with the NURETH-14 conference in Toronto, Canada.
 4. September 25-30, 201 - Special sessions on the OECD/NRC PSBT and BFBT Benchmarks, NURETH-14 conference, Toronto, Canada.
 5. November 2011 - NUREG report (draft) to be submitted to US NRC.
 6. December 2011 - OECD/NEA reports to be submitted.
30. The third workshop of the OECD/NRC PSBT Benchmark (PSBT-3) will be held in conjunction with the NURETH-14 conference in Toronto, Canada. The objectives of the PSBT-3 workshop will be:
 - discussion on final results of Phases I and II of the PSBT benchmark;
 - discussion on the progress with the OECD/NEA and NUREG reports;
 - discussion on the progress with the Special Journal Issue.
 31. Three technical reports on the benchmark results will be prepared:
 - NUREG report (US NRC) entitled “OECD/NEA/NRC Benchmark Based on NUPEC PWR Subchannel and Bundle Tests (PSBT): Experimental Database, Problem Specifications, and Final Results”;
 - two OECD/NEA reports entitled “OECD/NEA/NRC Benchmark Based on NUPEC PWR Subchannel and Bundle Tests (PSBT), Volume II: Benchmark Results for the Void Distribution Phase” and “OECD/NEA/NRC Benchmark Based on NUPEC PWR Subchannel and Bundle Tests (PSBT), Volume III: Benchmark Results for the Departure from Nucleate Boiling Phase”.

32. A special journal issue in the Journal of Science and Technology of Nuclear Installations is planned for the OECD/NRC PSBT benchmark. Editors of the special issue will be David Novog (McMaster University, Canada), Diana Cuervo (University Polytechnic of Madrid, Spain), Annalisa Manera (Paul Scherrer Institut, Switzerland), Alessandro Petruzzi (University of Pisa, Italy), and Maria Avramova (Coordinator of OECD/NRC PSBT benchmark).

APPENDIX 1

OECD/NRC Benchmark based on NUPEC PWR Sub-channel and Bundle Tests (PSBT) – Second Workshop (PSBT-2)

Hosted by
the Royal Institute of Technology (Kungliga Tekniska Högskolan, KTH)
Stockholm, Sweden

11-12 April 2011

AGENDA

Day 1: 11 April 2011 - Morning Sessions - Room FB53; Afternoon Sessions - Room FB42

Opening Session – Chair M. Avramova

- 09:00-09:10 Introduction and opening remarks - KTH
- 09:10-09:20 OECD/NEA presentation – J. Gulliford [P203]
- 09:20-09:30 Overview and status of PSBT benchmark activities after PSBT-1, M. Avramova [P204]

Technical Sessions on Phase I – Void Distribution Benchmark

Session I – Chair Y. Sung

- 09:30-09:50 Discussion on modifications and additional data included in the final benchmark specification, A. Rubin, M. Avramova [P205]
- 09:50-10:00 Presentation of support studies performed by the benchmark team, A. Rubin, M. Avramova [P206]

10:00-10:20 Coffee Break

- 10:20-10:50 Comparative analysis of submitted participants' results for Exercise I-1, A. Rubin, M. Avramova [P207]
- 10:50-11:20 Comparative analysis of submitted participants' results for Exercise I-2, A. Rubin, M. Avramova [P208]

Session II – Chair D. Cuervo

- 11:20-11:40 Exercise I-1 – Steady-state single sub-channel benchmark: Final results with porous media code THYC-COEUR, Franck Maurel [P209]
- 11:40-12:00 CFD analysis of boiling flow in PWR subchannel geometry using ANSYS CFX”, Johannes Weis [P210]
- 12:00-12:20 Numerical simulations of the steady-state subchannel test-case with NEPTUNE_CFD, C. Baudry, M. Guingo, A. Douce, J. Laviéville, S. Mimouni, M. Boucker [P211]

12:20-12:40 CFD analysis on the NUPEC PWR Subchannel and Bundle Test (PSBT), Eckhard Krepper, Roland Rzehak [P212]

12:40-13:00 CFD Simulation of BFBT and PSBT Experiments, C. N. Sökmén [P213]

13:00-14:00 Lunch

Session III – Chair E. Krepper

14:10-14:30 PSBT - CFD Simulation of Subcooled Boiling in Heated Subchannel Geometry using ANSYS CFX, Th. Frank, F. Reiterer, C. Lifante [P214]

14:30-14:50 Results of CFD analysis of PSBT Exercise-1 obtained by using STAR-CD, Victor Petrov, Simon Lo, Annalisa Manera [P215]

14:50-15:10 Exercise I-2 – Steady-state bundle benchmark Final Results with porous media code THYC-COEUR, Franck Maurel [P216]

15:10-15:30 Coffee Break

Session IV – Chair L. Sabotinov

15:30-16:00 Comparative analysis of submitted participants' results for Exercise I-3, A. Rubin, M. Avramova [P217]

16:00-16:10 Comparative analysis of submitted participants' results for Exercise I-4, A. Rubin, M. Avramova [P218]

16:10-16:30 Analysis of the NUPEC PSBT void distribution exercises I-1, I-2 and I-3 with FLICA-OVAP, M. Bucci, P. Fillion [P219]

16:30-16:50 Exercise I-3 – Transient bundle benchmark Final Results with porous media code THYC-COEUR, Franck Maurel [P220]

16:50-17:10 Transient bundle simulations by SUBCHANFLOW (Phase I, Exercise 3), U. Imke, V. Sanchez, M. Avramova [P221]

17:10-17:30 PSBT Test Bundle Modeling with STARCCM+, Emilian Popov, Robert Salko [P222]

19:00 Dinner

Day 2: 12 April 2011 - Room FB52

Session V – Chair M. Guingo

08:30-08:50 Void Fraction Prediction of NUPEC PSBT Tests by CATHARE Code, L. Michelotti, A. Del Nevo, F. D'Auria, F. Fiori [P223]

08:50-09:10 Analysis of PSBT Phase-I using FLICA4 and TRACE, Tae-Wan Kim, Annalisa Manera [P224]

09:10-09:30 Void fraction prediction with CATHARE 2, Luben Sabotinov [P225]

09:30-09:50 PSBT Benchmark and TRACE Model comparison for Phase-I, Exercise-2 & 3, Alexander Velazquez-Lozada [P226]

09:50-10:10 Coffee Break

Technical Sessions on Phase 2 – Departure from Nucleate Boiling Benchmark**Session VI – Chair T. Kozlowski**

- 10:10-10:40 Comparative analysis of submitted participants' results for Exercise II-1, A. Rubin, M. Avramova [P227]
- 10:40-11:00 Exercise II-1 – Steady-state temperature benchmark: Use of temperature measurements to calibrate turbulent mixing grid factor (porous media code THYC-COEUR), Franck Maurel [P228]
- 11:00-11:20 Models and results of CATHARE 3 simulations for PSBT phase II Exercise 1, Michel Valette [P229]
- 11:20-11:40 Comparative analysis of submitted participants' results for Exercise II-2, A. Rubin, M. Avramova [P230]

Session VII – Chair H. Gabriel

- 11:40-12:00 Exercise II-2 – Steady-state DNB benchmark Preliminary results with porous media code THYC-COEUR, Franck Maurel [P231]
- 12:00-12:20 Models and results of CATHARE 3 simulations for PSBT phase II Exercise 2, Michel Valette [P232]
- 12:20-12:40 Comparative analysis of submitted participants' results for Exercise II-3, A. Rubin, M. Avramova [P233]
- 12:40-13:00 First results of the NUPEC PSBT DNB exercises II-1, II-2 and II-3 with FLICA-OVAP", M. Bucci, P. Fillion [P234]

13:00-14:00 Lunch

Session VIII – Chair M. Valette

- 14:00-14:20 Analysis of PSBT Phase-I & II Exercises using a subchannel code MATRA, Phase II, K. W. Seo, S. J. Kim, and D. H. Hwang [P235]
- 14:20-14:40 Contribution of UNIPI to PSBT-2 Benchmark, F. Fiori, A. Kovtonyuk, A. Petruzzi [P236]
- 14:40-15:00 Assert VBRI Predictions for Phase 2 of the PSBT Benchmark, K. Leung, D. Novog [P237]
- 15:00-15:20 Preliminary Assessment of Improved ASSERT-PV CHF Models against PSBT Steady-State DNB Benchmark, Y. Rao, Z. Cheng, K. Leung, D. Novog [P238]

15:20-15:40 Coffee Break

Session IX – Chair D. Novog

- 15:40-16:00 BFBT transient void and pressure drop analysis with VIPRE-W, J-M. Le Corre, P. Alvarez, C. Adamsson [P239]
- 16:00-16:20 BFBT transient dryout analysis with VIPRE-W/MEFISTO-T, J-M. Le Corre, C. Adamsson [P240]
- 16:20-16:40 Action items and schedule of PSBT benchmark activities, M. Avramova [P241]
- 16:40 Conclusions and closing remarks, J. Gulliford

APPENDIX 2

OECD/NRC Benchmark based on NUPEC PWR Sub-channel and Bundle Tests (PSBT) – Second Workshop (PSBT-2)

Hosted by
Royal Institute of Technology (Kungliga Tekniska Högskolan, KTH)
Sweden
April 11-12, 2011

LIST OF PARTICIPANTS

Participants identified by * could not attend the workshop but have requested a Summary record

CANADA

Matthew Ball	ballmr@mcmaster.ca
Matt Krause	krausem@aecl.ca
Ken Leung	leungk4@mcmaster.ca
David Novog	novog@mcmaster.ca
Yanfei Rao	raoy@aecl.ca

CZECH REPUBLIC

Jan Hádek *	hdj@ujv.cz
Milan Malacka	mal@ujv.cz

FRANCE

Cyril Baudry	cyril.baudry@edf.fr
Matteo Bucci	matteo.bucci@cea.fr
Mathieu Guingo	mathieu.guingo@edf.fr
Philippe Fillion	philippe.fillion@cea.fr
Franck Maurel	franck.maurel@edf.fr
Luben Sabotinov	luben.sabotinov@irsn.fr
Enrico Sartori	esartori@noos.fr
Michel Valette	michel.valette@cea.fr

GERMANY

Thomas Frank	thomas.frank@ansys.com
Harry Gabriel	harry.gabriel@areva.com
Uwe Imke *	uwe.imke@kit.edu
Sergiy Khodyachykh	sergiy.khodyachykh@siemens.com
Markus Klein *	markus.klein@grs.de
Hans Koopman	hans.koopman@siemens.com
Eckhard Krepper	e.krepper@hzdr.de
Ihor Pasichnyk *	ihor.pasichnyk@grs.de
Johannes Weis	johannes.weis@grs.de

HUNGARY

András Keresztúri *	kere@aeki.kfki.hu
István Panka	panka@aeki.kfki.hu

ITALY

Filippo Fiori	filippo.fiori@for.unipi.it
Alessandro Del Nevo *	alessandro.delnevo@enea.it
Carlo Parisi *	carlo.parisi@enea.it
Alessandro Petruzzi *	a.petruzzi@ing.unipi.it

JAPAN

Ryosuke Nakamura *	ryo-naka@nfi.co.jp
Naofumi Tsukamoto *	n-tsuka@nfi.co.jp
Hideaki Utsuno *	utsuno-hideaki@jnes.go.jp

KOREA

Seo Kyong-Won	nulmiso@kaeri.re.kr
---------------	---------------------

RUSSIAN FEDERATION

Grigory Alekhin *	alechin@grpress.podolsk.ru
-------------------	----------------------------

SPAIN

Diana Cuervo	d.cuervo@upm.es
--------------	-----------------

SWEDEN

Ivan Gajev *	gajev@safety.sci.kth.se
Jean-Marie Le Corre	lecorrm@westinghouse.com

SWITZERLAND

Hakim Ferroukhi	hakim.ferroukhi@psi.ch
Tae-Wan Kim	taewan.kim@psi.ch
Annalisa Manera	annalisa.manera@psi.ch

TURKEY

Cemal Niyazi Sökmen	cemalniyazi.sokmen@hacettepe.edu.tr
---------------------	-------------------------------------

UNITED KINGDOM

Simon Lo	simon.lo@uk.cd-adapco.com
----------	---------------------------

UNITED STATES OF AMERICA

Thomas Downar	downar@umich.edu
Matthew Jessee	jesseema@ornl.gov
Emilian Popov	popove@ornl.gov
Adam Rubin	ajr5052@psu.edu
Yixing Sung	sungy@westinghouse.com
Mark Williams	williamsml@ornl.gov
Aaron Wysocki	awysock@umich.edu

OECD

Jim Gulliford	jim.gulliford@oecd.org
---------------	------------------------

INTERNATIONAL

Maria Avramova	mna109@psu.edu
Kostadin Ivanov	kni1@psu.edu
Tomasz Kozlowski	tomasz@safety.sci.kth.se
Kiril Velkov *	kiril.velkov@grs.de